

1. The Fourier series of a periodic waveform is

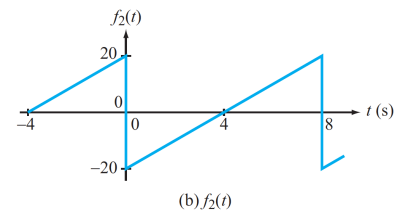
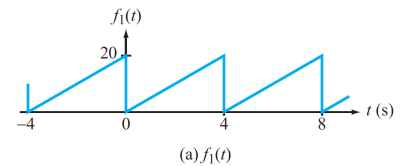
$$f_1(t) = 10 - \frac{20}{\pi} \sum_{n=1}^{\infty} \left(\frac{1}{n} \sin\left(\frac{n\pi t}{2}\right) \right)$$

(a) (20 pts) Determine the Fourier series of waveform $f_2(t)$.

(b) (10 pts) Determine the total average power of $f_2(t)$.

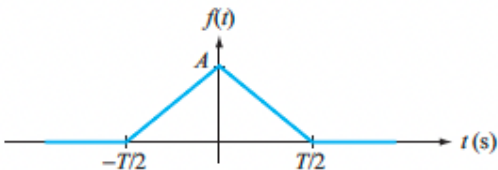
(Hint: $\sum_{n=1}^{\infty} \left(\frac{1}{n^2} \right) = \left(\frac{\pi^2}{6} \right)$)

(c) (10 pts) Determine the fractional power of the second harmonic with respect to the total average power.

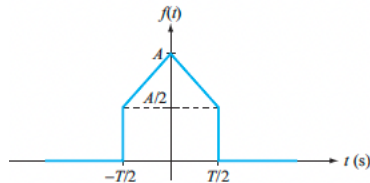


2. Determine the Fourier transform of the following waveforms:

(a) (30 pts) Assume $A = 10$ and $T = 6$ s



(b) (30 pts) Assume $A = 1$ and $T = 2$ s



1. The Fourier series of a periodic waveform is

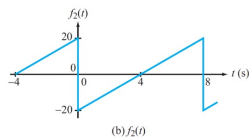
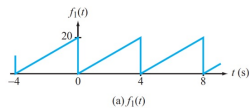
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a) $f_2(t) = 2 f_1\left(\frac{t}{2}\right) - 20$

$$2 f_1(t) = 2 \left[10 - \frac{20}{\pi} \sum_{n=1}^{\infty} \left(\frac{1}{n} \sin\left(\frac{n\pi t}{2}\right) \right) \right] = 20 - \frac{40}{\pi} \sum_{n=1}^{\infty} \left(\frac{1}{n} \sin\left(\frac{n\pi t}{2}\right) \right)$$

$$2 f_1(t) - 20 = -\frac{40}{\pi} \sum_{n=1}^{\infty} \left(\frac{1}{n} \sin\left(\frac{n\pi t}{2}\right) \right)$$

$$2 f_1\left(\frac{t}{2}\right) - 20 = -\frac{40}{\pi} \sum_{n=1}^{\infty} \left(\frac{1}{n} \sin\left(\frac{n\pi t}{4}\right) \right)$$

$$f_2(t) = -\frac{40}{\pi} \sum_{n=1}^{\infty} \left(\frac{1}{n} \sin\left(\frac{n\pi t}{4}\right) \right)$$

b) $P_x = \sum_{n=1}^{\infty} \left(\frac{40}{\pi} \right)^2 \left(\frac{1}{2n^2} \right)$

$$b = -\frac{40}{\pi}$$

$$P_x = \frac{800}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \Rightarrow \frac{800}{\pi^2} \left(\frac{\pi^2}{6} \right) = \frac{400}{3} \approx 133.33 \text{ [W]}$$

c.) fractional power = $\frac{P_2}{P_{av}}$

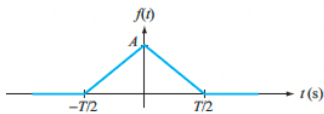
$$P_2 = \frac{b^2}{2} = \frac{200}{\pi^2}$$

$$P_{av} = \frac{400}{3}$$

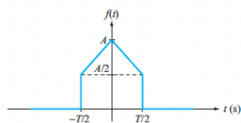
$$\text{fraction} = 15.2\%$$

2. Determine the Fourier transform of the following waveforms:

(a) (30 pts) Assume $A = 10$ and $T = 6s$



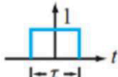
(b) (30 pts) Assume $A = 1$ and $T = 2s$



a) $\frac{AT}{2} \text{sinc}^2\left(\frac{\omega T}{4}\right)$

$$F(\omega) = \frac{60}{2} \text{sinc}^2\left(\frac{6\omega}{4}\right)$$

$$F(\omega) = 30 \text{sinc}^2\left(\frac{6\omega}{4}\right)$$

5.  $\text{rect}(t/\tau) \longleftrightarrow \tau \text{sinc}(\omega\tau/2)$

b) rectangle =

$$\text{Height} = \frac{A}{2} = \frac{1}{2}, \text{ Width} = T = 2$$

$$= \text{sinc}(\omega)$$

$$\text{Triangle} = \frac{1}{2} \text{sinc}^2\left(\frac{\omega}{2}\right)$$

$$F(\omega) = \text{sinc}(\omega) + \frac{1}{2} \text{sinc}^2\left(\frac{\omega}{2}\right)$$